Matrix-Free PCG for the Mode-k RKHS Subproblem with Kronecker Structure, Preconditioning, Scaling, and Concentration Justification

1 Problem statement and dimensions

Fix a tensor of order d with mode sizes $n_1, \ldots, n_d$. Let $k \in \{1, \ldots, d\}$ be the RKHS-regularized mode. Define

$$n := n_k, \qquad M := \prod_{\ell \neq k} n_\ell, \qquad N := nM = \prod_{\ell=1}^{d} n_\ell.$$

Assume we have a set of observed entries in the mode-k unfolding of size q with

$$n, r < q \ll N.$$

Let r be the CP rank (number of components).

Let $K \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite (PSD) kernel matrix (RKHS Gram matrix). We will allow the singular case $K \succeq 0$ and introduce a stabilization nugget $K_\epsilon := K + \epsilon I$ as needed.

Let $Z \in \mathbb{R}^{M \times r}$ denote the Khatri–Rao product of the factor matrices in all modes except k:

$$Z = A_d \odot \cdots \odot A_{k+1} \odot A_{k-1} \odot \cdots \odot A_1,$$

but *critically* Z must never be formed explicitly (its size is $M \times r$ with M potentially enormous).

Let $S \in \mathbb{R}^{N \times q}$ be a selection matrix that extracts the q observed entries from an N-vector, and set

$$P := SS^\mathsf{T} \in \mathbb{R}^{N \times N}.$$

Then P is a symmetric idempotent projector onto the observed coordinates.

The subproblem is to solve the linear system

$$\Big[(Z \otimes K)^\mathsf{T} P (Z \otimes K) + \lambda (I_r \otimes K)\Big] \operatorname{vec}(W) = (I_r \otimes K) \operatorname{vec}(B), \tag{1}$$

where $\lambda > 0$, $W \in \mathbb{R}^{n \times r}$ is the unknown, and $B \in \mathbb{R}^{n \times r}$ is a known right-hand-side matrix arising from the residual/projection step of the overall CP/RKHS method.

Goal. We must solve (1) by *matrix-free* PCG using only:

- kernel multiplies/solves with K (or K_ϵ),
- access to the observation index set Ω of size q,
- factor matrices A_ℓ (for $\ell \neq k$) to compute rows of Z implicitly,
- $O(nr + r^2)$ working memory (plus factor storage),

and avoid any cost that scales with N or any object of length N or matrix of shape $M \times r$.

2 Core identities and the masking operator

2.1 Vec–Kronecker identity

Lemma 1 (Vec–Kronecker identity)**.** *For compatible matrices A, B, X,*

$$(A \otimes B)\ \mathrm{vec}(X) = \mathrm{vec}(BXA^{\mathsf{T}}).$$

Proof. This is standard and follows from the definition of the Kronecker product and vectorization ordering. □

2.2 Selection mask as an entrywise projector

We work with the mode-k unfolding of shape $n \times M$. Let $\Omega \subset [n] \times [M]$ be the set of observed indices in this unfolding, with $|\Omega| = q$. We represent $P = SS^{\mathsf{T}}$ as the projector acting on $\mathrm{vec}(\cdot)$ that keeps exactly those entries and zeros the rest.

Define the masking operator $\mathcal{P}_\Omega : \mathbb{R}^{n\times M} \to \mathbb{R}^{n\times M}$ by

$$(\mathcal{P}_\Omega(X))_{ij} = \begin{cases} X_{ij}, & (i,j) \in \Omega, \\ 0, & \text{otherwise.} \end{cases}$$

Then, by construction of S,

$$P\ \mathrm{vec}(X) = \mathrm{vec}(\mathcal{P}_\Omega(X)). \tag{2}$$

Remark 1. The matrix $P \in \mathbb{R}^{N\times N}$ is never formed. The operator $\mathcal{P}_\Omega$ is applied via the index list Ω using scatter/gather (Section 4).

3 Exact matrix-free form of the linear operator

Define the system matrix

$$A := (Z \otimes K)^{\mathsf{T}} P (Z \otimes K) + \lambda (I_r \otimes K), \qquad b := (I_r \otimes K)\,\mathrm{vec}(B).$$

3.1 Derivation of the operator action in matrix form

Lemma 2 (Matrix form of $A\,\mathrm{vec}(W)$)**.** *Let $W \in \mathbb{R}^{n\times r}$. Then*

$$A\ \mathrm{vec}(W) = \mathrm{vec}\Big(K\big(\mathcal{P}_\Omega(KWZ^{\mathsf{T}})\,Z\big) + \lambda\,KW\Big). \tag{3}$$

Proof. Using Lemma 1,

$$(Z \otimes K)\,\mathrm{vec}(W) = \mathrm{vec}(KWZ^{\mathsf{T}}).$$

Apply the mask via (2):

$$P(Z \otimes K)\,\mathrm{vec}(W) = P\,\mathrm{vec}(KWZ^{\mathsf{T}}) = \mathrm{vec}(\mathcal{P}_\Omega(KWZ^{\mathsf{T}})).$$

Now apply $(Z \otimes K)^{\mathsf{T}} = Z^{\mathsf{T}} \otimes K$ and Lemma 1 again:

$$(Z^{\mathsf{T}} \otimes K)\ \mathrm{vec}(X) = \mathrm{vec}(KXZ),$$

so with $X = \mathcal{P}_\Omega(KWZ^{\mathsf{T}})$,

$$(Z \otimes K)^{\mathsf{T}} P (Z \otimes K)\,\mathrm{vec}(W) = \mathrm{vec}\Big(K\big(\mathcal{P}_\Omega(KWZ^{\mathsf{T}})\,Z\big)\Big).$$

Finally,

$$\lambda(I_r \otimes K)\,\mathrm{vec}(W) = \lambda\,\mathrm{vec}(KW),$$

and adding yields (3). □

3.2 SPD property and applicability of CG/PCG

Proposition 1 (SPD)**.** *Assume $K \succ 0$ and $\lambda > 0$. Then $A \succ 0$ is symmetric positive definite, hence CG/PCG applies. If $K \succeq 0$ is singular, then for any $\epsilon > 0$, the stabilized $K_\epsilon := K + \epsilon I \succ 0$ yields an SPD system with K replaced by K_ϵ.*

Proof. Symmetry is immediate since P is symmetric, K is symmetric, and Kronecker products preserve symmetry.

For definiteness, let $x \in \mathbb{R}^{nr}$, $x \neq 0$. Write $x = \operatorname{vec}(W)$. Then

$$x^\mathsf{T}(Z \otimes K)^\mathsf{T} P (Z \otimes K) x = \|(P^{1/2})(Z \otimes K)x\|_2^2 \geq 0$$

since P is an orthogonal projector ($P^{1/2} = P$). Thus the first term is PSD.

The regularization term satisfies

$$x^\mathsf{T}(\lambda I_r \otimes K)x = \lambda \operatorname{vec}(W)^\mathsf{T}(I_r \otimes K)\operatorname{vec}(W) = \lambda \operatorname{tr}(W^\mathsf{T} K W).$$

If $K \succ 0$ and $W \neq 0$, then $\operatorname{tr}(W^\mathsf{T} K W) > 0$, hence the regularizer is strictly positive. Therefore the sum is SPD.

If $K \succeq 0$ is singular, replacing K by $K_\epsilon = K + \epsilon I \succ 0$ yields the same argument. □

4 Matrix-free matvec via the observation set ΩOmega

Lemma 2 reduces the matvec $Y = \operatorname{MatVec}(W)$ to computing

$$U := KW \in \mathbb{R}^{n \times r}, \qquad V := \mathcal{P}_\Omega(UZ^\mathsf{T})\, Z \in \mathbb{R}^{n \times r}, \qquad Y = KV + \lambda U.$$

The expensive object $UZ^\mathsf{T} \in \mathbb{R}^{n \times M}$ must never be formed; we compute V by a loop over Ω.

4.1 Scatter-add formula

Let $\Omega = \{(i_t, j_t)\}_{t=1}^q$ (1-based indices with $i_t \in [n]$, $j_t \in [M]$). Define $z_j \in \mathbb{R}^r$ as the j-th row of Z. Then

$$(UZ^\mathsf{T})_{i_t, j_t} = \langle U_{i_t,:}, z_{j_t} \rangle .$$

Masking keeps only entries in Ω, so

$$\Big(\mathcal{P}_\Omega(UZ^\mathsf{T})\Big)_{ij} = \begin{cases} \langle U_{i,:}, z_j \rangle, & (i,j) \in \Omega, \\ 0, & \text{else.} \end{cases}$$

Thus the product with Z accumulates:

$$V_{i,:} = \sum_{(i,j)\in\Omega} \big(\langle U_{i,:}, z_j \rangle \big) z_j. \tag{4}$$

This is computed by initializing $V = 0$ and, for each observed pair (i_t, j_t), computing

$$s_t := \langle U_{i_t,:}, z_{j_t} \rangle, \qquad V_{i_t,:} \mathrel{+}= s_t\, z_{j_t}.$$

Total arithmetic for the loop is $O(qr)$ plus the cost to obtain each z_{j_t} implicitly (Section 5).

4.2 Row-count scaling for nonuniform sampling

If sampling over rows $i \in [n]$ is nonuniform, it is advantageous to reweight residuals by row counts to reduce anisotropy.

Let

$$c_i := \#\{t : i_t = i\}, \qquad i = 1, \dots, n,$$

computed in $O(q)$ time. For a damping $\tau \geq 0$, define

$$D := \operatorname{diag}\left(\frac{1}{\sqrt{c_i + \tau}}\right)_{i=1}^{n}.$$

Define a weighted mask operator $\mathcal{P}_{\Omega,D}$ by multiplying each observed entry (i_t, j_t) by $d_{i_t}^2$:

$$(\mathcal{P}_{\Omega,D}(X))_{i_t,j_t} := d_{i_t}^2 X_{i_t,j_t}, \quad (\mathcal{P}_{\Omega,D}(X))_{ij} = 0 \text{ otherwise.}$$

Equivalently, at the vectorized level,

$$P_D := (D \otimes I_M)\, P\, (D \otimes I_M), \qquad P_D \operatorname{vec}(X) = \operatorname{vec}(\mathcal{P}_{\Omega,D}(X)).$$

The same derivation as Lemma 2 holds with P replaced by P_D.

In practice, (4) becomes

$$V_{i,:} = \sum_{(i,j)\in\Omega} d_i^2 \big(\langle U_{i,:}, z_j \rangle \big) z_j, \tag{5}$$

i.e. the scalar s_t is replaced by $d_{i_t}^2 s_t$.

5 Implicit computation of rows of ZZ without forming ZZ

5.1 Khatri–Rao row formula

Let factor matrices for $\ell \neq k$ be $A_\ell \in \mathbb{R}^{n_\ell \times r}$. The Khatri–Rao product $Z = \odot_{\ell \neq k} A_\ell$ has rows indexed by a multi-index $(i_\ell)_{\ell \neq k}$, and

$$z_{(i_\ell)} = a_{i_1}^{(1)} \circ \cdots \circ a_{i_{k-1}}^{(k-1)} \circ a_{i_{k+1}}^{(k+1)} \circ \cdots \circ a_{i_d}^{(d)} \in \mathbb{R}^r,$$

where $a_i^{(\ell)}$ denotes row i of A_ℓ.

5.2 Mixed-radix decoding from a single column index

We must map an index $j \in \{1, \dots, M\}$ to its multi-index $(i_\ell)_{\ell \neq k}$ deterministically, without precomputing a size-M map.

Fix an ordering of the modes excluding k:

$$(\ell_1, \dots, \ell_{d-1}) = (1, 2, \dots, k-1, k+1, \dots, d).$$

Let their sizes be n_{ℓ_m} and define $M = \prod_{m=1}^{d-1} n_{\ell_m}$.

We use 0-based $j_0 := j - 1 \in \{0, \dots, M-1\}$. Define strides (little-endian / column-major):

$$s_1 = 1, \qquad s_m = \prod_{t=1}^{m-1} n_{\ell_t} \quad (m \geq 2).$$

Then decode:

$$i_{\ell_m} = 1 + \left(\left\lfloor \frac{j_0}{s_m} \right\rfloor \bmod n_{\ell_m} \right), \qquad m = 1, \dots, d-1. \tag{6}$$

This is exact mixed-radix representation of j_0.

5.3 ImplicitRowZ procedure and cost

Given j, compute indices via (6), then compute

$$z \leftarrow a^{(\ell_1)}_{i_{\ell_1}} \in \mathbb{R}^r, \quad \text{for } m = 2, \dots, d-1 : \; z \leftarrow z \circ a^{(\ell_m)}_{i_{\ell_m}}.$$

This uses $(d-2)$ Hadamard products of length-r vectors: arithmetic $O((d-2)r)$ and memory $O(r)$.

Remark 2 (Caching). If many observations share the same column index j (repeated fibers/columns in Ω), caching computed z_j can reduce the total time from $q \cdot O((d-2)r)$ to $|\{j_t\}| \cdot O((d-2)r)$ at the cost of storing up to $|\{j_t\}|$ vectors of length r.

6 Preconditioners: Kronecker surrogates, scaling, and application

6.1 Computing $G = Z^\top Z$**G=Z**$^T Z without forming$**Z**Z

A crucial identity for Khatri–Rao products:

Lemma 3 (Khatri–Rao Gram identity)**.** *Let $Z = A_{\ell_{d-1}} \odot \cdots \odot A_{\ell_1}$ with each $A_{\ell_m} \in \mathbb{R}^{n_{\ell_m} \times r}$. Then*

$$Z^\top Z = (A^\top_{\ell_{d-1}} A_{\ell_{d-1}}) \circ \cdots \circ (A^\top_{\ell_1} A_{\ell_1}).$$

Proof. Entrywise, for $p, q \in [r]$,

$$(Z^\top Z)_{pq} = \sum_{\text{rows } t} Z_{tp} Z_{tq}.$$

Each row t corresponds to a multi-index, and

$$Z_{tp} Z_{tq} = \prod_{m=1}^{d-1} (A_{\ell_m})_{i_{\ell_m},p} \, (A_{\ell_m})_{i_{\ell_m},q}.$$

Summing over the Cartesian product of indices factorizes into a product of sums, yielding the Hadamard product of Gram matrices. □

Thus we compute $G := Z^\top Z \in \mathbb{R}^{r \times r}$ in

$$O\left(r^2 \sum_{\ell \neq k} n_\ell\right)$$

time and $O(r^2)$ memory, without forming Z.

6.2 Base Kronecker preconditioner

Motivated by separable approximations of the data term (Sections 9–10), define

$$\mathcal{M} := (G + \lambda I_r) \otimes K_\epsilon, \qquad K_\epsilon := K + \epsilon I, \; \epsilon \geq 0.$$

Proposition 2 (SPD and invertibility)**.** *If $\lambda > 0$ and $K_\epsilon \succ 0$, then $\mathcal{M} \succ 0$ and*

$$\mathcal{M}^{-1} = (G + \lambda I_r)^{-1} \otimes K_\epsilon^{-1}.$$

Proof. $G \succeq 0$ and $\lambda > 0$ imply $G + \lambda I_r \succ 0$. Kronecker product of SPD matrices is SPD and invertible, with inverse equal to the Kronecker product of inverses. □

6.3 Applying the preconditioner inverse in matrix form

Given $R \in \mathbb{R}^{n \times r}$, to compute X satisfying $\mathrm{vec}(X) = \mathcal{M}^{-1}\,\mathrm{vec}(R)$:

$$X = K_\epsilon^{-1} R (G + \lambda I_r)^{-1}.$$

Implementation uses:

- a Cholesky factorization $K_\epsilon = LL^\mathsf{T}$ (once per outer step),
- a Cholesky factorization $G + \lambda I_r = RR^\mathsf{T}$ (once per outer step),

then triangular solves for the r right-hand sides.

6.4 Mask-aware similarity-scaled preconditioner with row scaling

With row scaling D from Section 4.2, define

$$\mathcal{M}_D := (G + \lambda I_r) \otimes (D^{-1} K_\epsilon D^{-1}).$$

This remains SPD if $K_\epsilon \succ 0$ and D has positive diagonal. Notably,

$$(D^{-1} K_\epsilon D^{-1})^{-1} = D\, K_\epsilon^{-1}\, D,$$

so applying $\mathcal{M}_D^{-1}$ to R is:

$$X = D\, K_\epsilon^{-1} (DR)\, (G + \lambda I_r)^{-1}.$$

This adds only $O(nr)$ diagonal scalings to the preconditioner application.

6.5 Adaptive scalar scaling αalpha via an energy-matching probe

In practice, the data term magnitude depends on sampling and mask pattern. Introduce a scalar $\alpha > 0$ in the preconditioner:

$$\mathcal{M}_\alpha := (\alpha G + \lambda I_r) \otimes K_\epsilon \quad (\text{or } \mathcal{M}_{D,\alpha} := (\alpha G + \lambda I_r) \otimes (D^{-1} K_\epsilon D^{-1})).$$

A principled scalar α is obtained by matching quadratic forms along a probe direction W_0:

$$\alpha := \frac{\big\langle W_0, K\big(\mathcal{P}_\Omega(K W_0 Z^\mathsf{T}) Z\big)\big\rangle_\mathrm{F}}{\big\langle W_0, K\big((K W_0) G\big)\big\rangle_\mathrm{F}}. \tag{7}$$

Here the numerator is one masked matvec data-term evaluation, and the denominator uses G only. This is a Rayleigh-quotient style scale match: it chooses α so that the separable surrogate aligns with the true masked data term in energy along W_0.

Remark 3. Compute α once per outer ALS sweep, or occasionally. Warm-starting with the previous W as W_0 often yields a stable and meaningful scale.

6.6 Fallback diagonal/block-Jacobi preconditioners (optional)

If forming/factorizing $G + \lambda I_r$ becomes expensive for large r, use

$$\mathcal{M}_\mathrm{diag} := (\mathrm{diag}(G) + \lambda I_r) \otimes K_\epsilon,$$

or even $(\alpha I_r) \otimes K_\epsilon$. These are weaker but cheaper; the same application logic holds with a diagonal right-solve.

7 PCG algorithm in matrix form

We solve $A\,\mathrm{vec}(W) = b$ but implement everything using $n \times r$ matrices and Frobenius inner products:

$$\langle X, Y \rangle_{\mathrm{F}} := \mathrm{tr}(X^{\mathsf{T}} Y).$$

This is exactly the Euclidean inner product of vectorizations: $\langle X, Y \rangle_{\mathrm{F}} = \langle \mathrm{vec}(X), \mathrm{vec}(Y) \rangle$.

7.1 MatVec routine

Given $W \in \mathbb{R}^{n \times r}$:

1. $U \leftarrow K_\epsilon W$ (kernel multiply).
2. Initialize $V \leftarrow 0 \in \mathbb{R}^{n \times r}$.
3. For each observed pair $(i_t, j_t) \in \Omega$:
   (a) compute $z \leftarrow z_{j_t}$ via implicit decoding and Hadamard products (Section 5),
   (b) compute scalar $s \leftarrow \langle U_{i_t,:}, z \rangle$,
   (c) if row scaling is used: $s \leftarrow d_{i_t}^2 s$,
   (d) accumulate $V_{i_t,:} \leftarrow V_{i_t,:} + s z^{\mathsf{T}}$.
4. $Y \leftarrow K_\epsilon V + \lambda U$.

Return Y.

7.2 Preconditioner solve routine

Given $R \in \mathbb{R}^{n \times r}$:

$$X = \mathcal{M}^{-1} R \quad \text{implemented as} \quad X = K_\epsilon^{-1} R (\alpha G + \lambda I_r)^{-1},$$

optionally with similarity scaling:

$$X = D\, K_\epsilon^{-1} (DR)\, (\alpha G + \lambda I_r)^{-1}.$$

7.3 PCG pseudocode

Algorithm 1 Matrix-free PCG for the mode-k RKHS system

Require: Observation list $\Omega = \{(i_t, j_t)\}_{t=1}^{q}$, factors $\{A_\ell\}_{\ell \neq k}$, kernel K (or K_ϵ), RHS matrix B, parameters (λ, ϵ), optional row scaling D, optional scalar α (or compute via probe)

Build $G = Z^\mathsf{T} Z$ via Hadamard Gram identity (Lemma 3)
Form $K_\epsilon \leftarrow K + \epsilon I$; factorize $K_\epsilon = LL^\mathsf{T}$
Choose α (either $\alpha = 1$ or via probe (7)); factorize $(\alpha G + \lambda I_r) = RR^\mathsf{T}$
Initialize W_0 (zeros or warm start)
$R_0 \leftarrow B - \mathrm{MatVec}(W_0)$ ▷ matrix residual
$Z_0 \leftarrow \mathrm{PrecSolve}(R_0)$
$P_0 \leftarrow Z_0$
$\gamma_0 \leftarrow \langle R_0, Z_0 \rangle_\mathrm{F}$
for $m = 0, 1, 2, \ldots$ until convergence **do**
    $Q_m \leftarrow \mathrm{MatVec}(P_m)$
    $\alpha_m \leftarrow \gamma_m / \langle P_m, Q_m \rangle_\mathrm{F}$
    $W_{m+1} \leftarrow W_m + \alpha_m P_m$
    $R_{m+1} \leftarrow R_m - \alpha_m Q_m$
    if stopping criterion satisfied **then**
        break
    end if
    $Z_{m+1} \leftarrow \mathrm{PrecSolve}(R_{m+1})$
    $\gamma_{m+1} \leftarrow \langle R_{m+1}, Z_{m+1} \rangle_\mathrm{F}$
    $\beta_m \leftarrow \gamma_{m+1} / \gamma_m$
    $P_{m+1} \leftarrow Z_{m+1} + \beta_m P_m$
end for
return W_{m+1}

7.4 Stopping criteria and numerics

Two standard stopping rules:

$$\frac{\|R_m\|_\mathrm{F}}{\|B\|_\mathrm{F}} \leq \mathrm{tol}, \quad \text{or} \quad \frac{\sqrt{\langle R_m, Z_m \rangle_\mathrm{F}}}{\sqrt{\langle R_0, Z_0 \rangle_\mathrm{F}}} \leq \mathrm{tol}.$$

Typical tolerances: $\mathrm{tol} \in [10^{-6}, 10^{-4}]$ depending on outer-loop accuracy needs.

Nugget choice. If K is ill-conditioned or singular, choose $\epsilon = \eta \, \mathrm{tr}(K)/n$ with $\eta \in [10^{-8}, 10^{-4}]$.

Warm start. Initialize W_0 from the previous outer iteration's W (or extrapolation) to reduce PCG iterations.

8 Complexity analysis: precomputation, per-iteration cost, and regimes

Define:

- $C_K(n,r)$ = cost of multiplying $K_\epsilon X$ for $X \in \mathbb{R}^{n\times r}$. For dense K_ϵ, $C_K = O(n^2 r)$; for sparse K_ϵ, $C_K = O(\mathrm{nnz}(K_\epsilon)\, r)$.
- $C_Z(d,r) = O((d-2)r)$ = cost to compute an implicit row z_j (Section 5).

8.1 Precomputation

- Compute Gram matrices $A_\ell^\top A_\ell$ for each $\ell \neq k$: total $O\Big(r^2 \sum_{\ell\neq k} n_\ell\Big)$.
- Compute G via Hadamard products: $O(r^2(d-1))$.
- Factorize $\alpha G + \lambda I_r$: $O(r^3)$.
- Factorize K_ϵ (dense): $O(n^3)$.
- Compute row counts c_i and optionally D: $O(q)$.

None of these depend on N.

8.2 Per-iteration matvec cost

Each matvec requires:

- $U = K_\epsilon W$: $C_K(n,r)$.
- Loop over q observations: per observation

  $$\text{ImplicitRowZ} : C_Z(d,r), \quad \text{dot+axpy} : O(r),$$

  so total $O(q(C_Z(d,r) + r)) = O(qdr)$.
- $K_\epsilon V$: $C_K(n,r)$.
- add λU: $O(nr)$.

Therefore

$$T_{\text{matvec}} = 2C_K(n,r) + O(qdr).$$

8.3 Per-iteration preconditioner cost

With Cholesky factors precomputed:

- solve $K_\epsilon Y = R$ for r RHS: dense $O(n^2 r)$ (triangular solves), sparse depends on factorization structure,
- right-solve $Y(\alpha G + \lambda I)^{-1}$: $O(nr^2)$,
- optional diagonal scalings by D: $O(nr)$.

Thus

$$T_{\text{prec}} = O(n^2 r + nr^2).$$

8.4 Total PCG iteration cost

Dominant terms (dense K_ϵ):

$$T_{\text{iter}} = O(n^2 r) + O(qdr) \quad \text{(plus smaller } O(nr^2)\text{)}.$$

Memory (excluding factor storage) is $O(nr + r^2)$ plus optional cache.

8.5 Regime guidance

- If $qdr \gg n^2 r$, the Ω-loop dominates: caching z_j for repeated j and pre-decoding indices are high impact.
- If $n^2 r \gg qdr$, kernel operations dominate: use structured/sparse kernels or optional low-rank approximations to reduce C_K.

9 Conditioning in an ideal separable surrogate model

This section justifies why Kronecker preconditioning is natural and yields explicit eigenvalue formulas in a separable approximation.

9.1 Separable surrogate

Conceptually define $X := Z \otimes K_\epsilon$ and consider the unmasked full Gram

$$X^\top X = (Z^\top Z) \otimes (K_\epsilon^\top K_\epsilon) = G \otimes K_\epsilon^2 \quad \text{(since } K_\epsilon \text{ symmetric)}.$$

A separable surrogate for the regularized operator is

$$\widetilde{A} := \alpha(G \otimes K_\epsilon^2) + \lambda(I_r \otimes K_\epsilon), \tag{8}$$

where $\alpha > 0$ encodes the effective sampling/data-term scale (estimated via probe (7) or conceptually $\alpha \approx q/N$ under uniform sampling).

9.2 Preconditioned surrogate eigenvalues

Use preconditioner

$$\mathcal{M}_\alpha := (\alpha G + \lambda I_r) \otimes K_\epsilon.$$

Then

Lemma 4 (Closed-form spectrum of $\mathcal{M}_\alpha^{-1}\widetilde{A}$)**.** *Let $G = U\Sigma U^\top$ with eigenvalues $\sigma_i \geq 0$ and $K_\epsilon = V\Lambda V^\top$ with eigenvalues $\lambda_j > 0$. Then the eigenvalues of $\mathcal{M}_\alpha^{-1}\widetilde{A}$ are*

$$\mu_{ij} = \alpha\frac{\sigma_i}{\alpha\sigma_i + \lambda}\lambda_j + \lambda\frac{1}{\alpha\sigma_i + \lambda}, \qquad i = 1, \ldots, r,\ j = 1, \ldots, n. \tag{9}$$

Proof. Compute

$$\begin{aligned}\mathcal{M}_\alpha^{-1}\widetilde{A} &= \big((\alpha G + \lambda I)^{-1} \otimes K_\epsilon^{-1}\big)\big(\alpha G \otimes K_\epsilon^2 + \lambda I \otimes K_\epsilon\big)\\ &= \alpha(\alpha G + \lambda I)^{-1} G \otimes K_\epsilon + \lambda(\alpha G + \lambda I)^{-1} \otimes I.\end{aligned}$$

In the basis $u_i \otimes v_j$, $(\alpha G + \lambda I)^{-1}G$ acts by $\frac{\sigma_i}{\alpha\sigma_i+\lambda}$ and K_ϵ acts by λ_j, while $(\alpha G + \lambda I)^{-1}$ acts by $\frac{1}{\alpha\sigma_i+\lambda}$. Summing yields (9). $\square$

Thus

$$\kappa(\mathcal{M}_\alpha^{-1}\widetilde{A}) = \frac{\max_{i,j}\mu_{ij}}{\min_{i,j}\mu_{ij}}.$$

This formula explicitly shows:

- the preconditioner removes one K_ϵ factor (reducing sensitivity to the K_ϵ^2 term),
- dependence on σ_i is moderated by the regularizer through $\alpha\sigma_i + \lambda$,
- increasing λ tends to stabilize and reduce spread in μ_{ij}.

10 Concentration justification: why the masked operator is close to a separable surrogate

We now provide the explicit matrix concentration reasoning discovered in iterations 6–7. The goal is to justify (under explicit assumptions) a PSD sandwich between the masked Gram and its expectation, which is separable.

10.1 Conceptual rows and Gram representation

Let $X := Z \otimes K_\epsilon \in \mathbb{R}^{N\times(nr)}$ (conceptual). Index rows by $t = (i,j) \in [n] \times [M]$; let the row be $x_t^\top \in \mathbb{R}^{nr}$. Then:

$$X^\top X = \sum_{t=1}^{N} x_t x_t^\top.$$

The observed mask Ω corresponds to selecting a subset of rows. For an unweighted mask (the actual case),

$$X^\top P X = \sum_{t\in\Omega} x_t x_t^\top.$$

10.2 With-replacement, importance-weighted model (cleanest concentration)

We first analyze i.i.d. sampling with replacement because it yields a clean sum of independent PSD matrices and standard Matrix Chernoff bounds.

Suppose $t_1, \ldots, t_q$ are i.i.d. samples from a distribution π on $\{1, \ldots, N\}$. Define PSD summands

$$Y_s := \frac{1}{q}\frac{1}{\pi(t_s)} x_{t_s} x_{t_s}^\top \succeq 0, \qquad s = 1, \ldots, q.$$

Define the empirical Gram

$$\widehat{\Sigma} := \sum_{s=1}^{q} Y_s.$$

Then

$$\mathbb{E}[\widehat{\Sigma}] = \sum_{t=1}^{N} x_t x_t^\top = X^\top X.$$

10.2.1 Bounded leverage condition

A sufficient condition for applying Matrix Chernoff is that each summand is uniformly bounded:

$$0 \preceq Y_s \preceq \frac{R}{q} I \quad \text{a.s.}$$

A sufficient and interpretable requirement is:

$$\frac{1}{\pi(t)} \|x_t\|_2^2 \le R \quad \forall t. \tag{10}$$

This is a leverage/row-norm control condition: no single row dominates after importance correction.

10.2.2 Matrix Chernoff template

Theorem 1 (Matrix Chernoff (template form))**.** *Let $Y_1, \dots, Y_q$ be independent PSD random matrices in dimension d such that $0 \preceq Y_s \preceq \frac{R}{q} I$ almost surely. Let $\Sigma := \mathbb{E}\big[\sum_{s=1}^{q} Y_s\big]$. Then for $0 < \delta < 1$,*

$$\Pr\left[\lambda_{\min}\Big(\sum_{s=1}^{q} Y_s\Big) \le (1-\delta)\lambda_{\min}(\Sigma)\right] \;\le\; d \exp\left(-c_1 \frac{\delta^2 \lambda_{\min}(\Sigma)}{R}\right),$$

$$\Pr\left[\lambda_{\max}\Big(\sum_{s=1}^{q} Y_s\Big) \ge (1+\delta)\lambda_{\max}(\Sigma)\right] \;\le\; d \exp\left(-c_2 \frac{\delta^2 \lambda_{\max}(\Sigma)}{R}\right),$$

for absolute constants $c_1, c_2 > 0$ (often $c_1 = \frac{1}{2}$ and $c_2 = \frac{1}{3}$ in standard statements). Thus with probability at least $1 - \eta$, choosing δ accordingly yields the PSD sandwich

$$(1-\delta)\Sigma \preceq \sum_{s=1}^{q} Y_s \preceq (1+\delta)\Sigma.$$

Applying Theorem 1 to $\widehat{\Sigma}$ yields:

$$(1-\delta)\, X^\top X \preceq \widehat{\Sigma} \preceq (1+\delta)\, X^\top X \quad \text{with high probability, under (10).} \tag{11}$$

10.3 Bridge to the actual mask: uniform sampling without replacement and unweighted sums

The real operator uses an unweighted without-replacement set Ω (a size-q subset). We now justify a closely analogous sandwich in that regime.

10.3.1 Expectation under uniform without replacement

Let Ω be uniformly distributed over all q-subsets of $\{1, \dots, N\}$. Then

$$\mathbb{E}\left[\sum_{t \in \Omega} x_t x_t^\top\right] = \frac{q}{N} \sum_{t=1}^{N} x_t x_t^\top = \frac{q}{N} X^\top X.$$

Thus the natural scale is $\alpha = q/N$. When N is huge, we do not compute this explicitly; we may estimate an equivalent scale α via (7).

10.3.2 Concentration without replacement (negative dependence principle)

Sampling without replacement exhibits negative dependence; for many bounded-sum quantities, it concentrates at least as well as with-replacement sampling. In matrix settings, one may invoke without-replacement variants of matrix Bernstein/Hoeffding inequalities, or use comparison principles showing that (for convex trace functionals or mgf-based arguments) without-replacement tails are bounded by with-replacement tails under matching marginals.

We therefore state the following explicit assumption-to-conclusion template.

Assumption 1 (Bounded row energy)**.** *There exists* $L > 0$ *such that*

$$\|x_t\|_2^2 \le L \quad \forall t \in \{1, \ldots, N\}.$$

Equivalently, $0 \preceq x_t x_t^\top \preceq LI$ *for all* t*.*

Theorem 2 (PSD sandwich for unweighted uniform without-replacement mask (template))**.** *Assume Assumption 1. Let* Ω *be uniform over all* q*-subsets of* $\{1, \ldots, N\}$ *and define*

$$\Sigma_\Omega := \sum_{t \in \Omega} x_t x_t^\top = X^\top P X.$$

Then for any $0 < \delta < 1$*, with probability at least* $1 - \eta$ *(for* η *depending on* (δ, q, d, L) *and spectral parameters of* $X^\top X$*),*

$$(1 - \delta)\frac{q}{N} X^\top X \preceq \Sigma_\Omega \preceq (1 + \delta)\frac{q}{N} X^\top X. \tag{12}$$

Remark 4. Theorem 2 is the without-replacement analog of (11). One may obtain explicit η using a named without-replacement matrix Bernstein/Hoeffding inequality; the key structural requirement is boundedness & adequate sample size q relative to dimension/intrinsic dimension. We keep the dependence symbolic to avoid baking in brittle constants; the logical role is to provide the PSD sandwich with an explicit error parameter δ.

10.4 Nonuniform sampling and how row scaling helps

If the observed set is not uniformly distributed, we parameterize marginal probabilities by a distribution π with

$$\pi_{\min} \le \pi(t) \le \pi_{\max}, \qquad \rho := \frac{\pi_{\max}}{\pi_{\min}} \ge 1.$$

When $\rho \approx 1$, sampling is near-uniform and (12) remains approximately valid after rescaling, with degradation that can be absorbed into an effective δ.

Row scaling D modifies rows by $x_{(i,j)} \mapsto \tilde{x}_{(i,j)} := d_i x_{(i,j)}$. This can reduce nonuniformity induced primarily by unequal row counts and thereby reduce ρ. This is precisely why Section 4.2 improves convergence in practice: it makes the effective sampling closer to uniform over the scaled design.

11 Final theorem template: PCG iteration bound under PSD sandwich and separable surrogate

Define the regularized operator

$$A := X^\top P X + \lambda(I_r \otimes K_\epsilon).$$

Under Theorem 2 (or the weighted i.i.d. model), we have with high probability:

$$(1-\delta)\alpha(G \otimes K_\epsilon^2) + \lambda(I \otimes K_\epsilon) \preceq A \preceq (1+\delta)\alpha(G \otimes K_\epsilon^2) + \lambda(I \otimes K_\epsilon), \tag{13}$$

where α is the expected sampling scale (conceptually $\alpha = q/N$, or estimated by probe).

Let $\widetilde{A}$ be as in (8). Then (13) implies

$$(1-\delta)\widetilde{A} \preceq A \preceq (1+\delta)\widetilde{A}.$$

Lemma 5 (PSD sandwich implies bounded preconditioned condition number)**.** *Let $\widetilde{A} \succ 0$ and suppose $(1-\delta)\widetilde{A} \preceq A \preceq (1+\delta)\widetilde{A}$ with $0 < \delta < 1$. Then for any SPD preconditioner $\mathcal{M} \succ 0$,*

$$\kappa(\mathcal{M}^{-1}A) \leq \kappa(\mathcal{M}^{-1}\widetilde{A}) \cdot \frac{1+\delta}{1-\delta}.$$

Proof. From $(1-\delta)\widetilde{A} \preceq A \preceq (1+\delta)\widetilde{A}$, pre- and post-multiply by $\mathcal{M}^{-1/2}$ to obtain

$$(1-\delta)\,\mathcal{M}^{-1/2}\widetilde{A}\mathcal{M}^{-1/2} \preceq \mathcal{M}^{-1/2}A\mathcal{M}^{-1/2} \preceq (1+\delta)\,\mathcal{M}^{-1/2}\widetilde{A}\mathcal{M}^{-1/2}.$$

Thus eigenvalues satisfy

$$(1-\delta)\lambda_i(\mathcal{M}^{-1}\widetilde{A}) \leq \lambda_i(\mathcal{M}^{-1}A) \leq (1+\delta)\lambda_i(\mathcal{M}^{-1}\widetilde{A}),$$

so

$$\frac{\lambda_{\max}(\mathcal{M}^{-1}A)}{\lambda_{\min}(\mathcal{M}^{-1}A)} \leq \frac{(1+\delta)\lambda_{\max}(\mathcal{M}^{-1}\widetilde{A})}{(1-\delta)\lambda_{\min}(\mathcal{M}^{-1}\widetilde{A})} = \kappa(\mathcal{M}^{-1}\widetilde{A}) \cdot \frac{1+\delta}{1-\delta}.$$

□

Theorem 3 (PCG convergence bound for the mode-k RKHS system (explicit template))**.** *Assume $K_\epsilon \succ 0$, $\lambda > 0$, and a PSD sandwich* (13) *holds with probability at least $1-\eta$ for some $\delta \in (0,1)$ and scale $\alpha > 0$. Let $\mathcal{M}_\alpha = (\alpha G + \lambda I_r) \otimes K_\epsilon$ (or $\mathcal{M}_{D,\alpha}$ with similarity scaling). Then with probability at least $1-\eta$,*

$$\kappa(\mathcal{M}_\alpha^{-1}A) \leq \kappa(\mathcal{M}_\alpha^{-1}\widetilde{A}) \cdot \frac{1+\delta}{1-\delta},$$

where $\widetilde{A}$ is the separable surrogate (8)*. Moreover, PCG iterates satisfy the standard error bound*

$$\|e_m\|_A \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^m \|e_0\|_A, \qquad \kappa := \kappa(\mathcal{M}_\alpha^{-1}A).$$

In the surrogate model, $\kappa(\mathcal{M}_\alpha^{-1}\widetilde{A})$ is determined exactly by eigenvalues (9)*.*

Remark 5. Theorem 3 is the rigorous articulation of the heuristic: if the mask behaves like a near-uniform sampler on the relevant design, then A is close to a separable operator, and the Kronecker preconditioner captures that structure, yielding fast PCG. Row scaling D improves near-uniformity and hence improves δ (and/or reduces nonuniformity ratio ρ).

12 Summary of the full discovered solution

Exactness of matvec. Lemma 2 and (4) show the matvec is exact and uses only Ω and implicit z_j.

No $O(N)$ costs. All computations avoid storing/iterating over N or forming Z; complexity is $O(n^2r) + O(qdr)$ per PCG iteration (dense K_ϵ).

Preconditioner practicality. G is computed without Z (Lemma 3), and $\mathcal{M}^{-1}$ is applied by solving with K_ϵ and an $r \times r$ SPD matrix.

Robustness. If K is singular/ill-conditioned, use nugget K_ϵ. If sampling is nonuniform, use row scaling D and optionally similarity-scaled preconditioning $\mathcal{M}_{D,\alpha}$.

Justification. Conditioning is explicit in the separable model (Lemma 4); closeness of the masked Gram to its expectation is justified under bounded-leverage and sufficient sampling by matrix concentration (Theorems 1 and 2), leading to Theorem 3.